



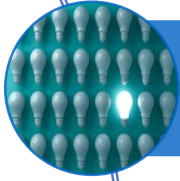
End-to-End Spectrometer Automatic Calibration Assistance Platform

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1. Background



2. Objectives



3. End-to-End Spectrometer Automatic Calibration Assistance Platform



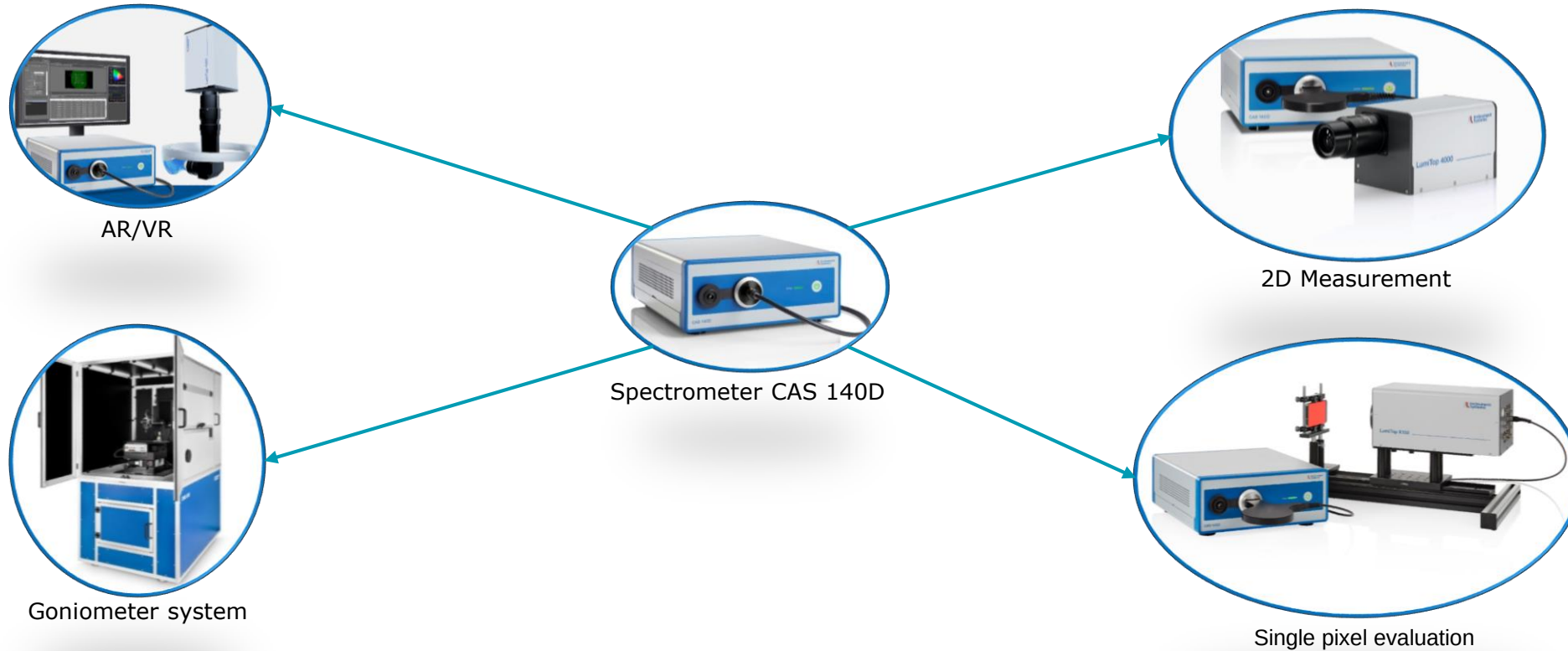
4. Results comparison



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1. Background

Spectrometer is the most important product of Instrument Systems Division (SISD). It is the core component of many of our equipment.



The spectrometer has very high accuracy, at same time it has relative fragile structure, therefore requires regular re-calibration to maintain its high precision.

As our sale grows, our calibration demand grows with it. This creates new challenges.

The challenges of spectrometer calibration came in 2 major categories:

The calibration composition:

The calibration is an extremely complicated process: involving >4 databases, >4 software, >5 reference standards, etc.

The calibration happens in many different sites in different countries, including both permanent sites (labs) and temporary sites (customers' factories).

Permanent sites: Shenzhen, Shanghai, Hanoi, etc.

Temporary sites: Zhengzhou, Longhua, Singapore, etc.

Many calibration sites are not supported by internal engineers, but by external engineers from our distributor partners.

Manufacturer Customers:

The calibration requests are not evenly distributed across the year, the requests are concentrated in 1-2 months in a year, normally before the production of new generation products.

To keep the production impact to the minimal, the requirement for calibration turnover rate is very high.

The calibration environment is not always desirable: like wearing dust-proof clothing in clean-room to do calibration is not pleasant.

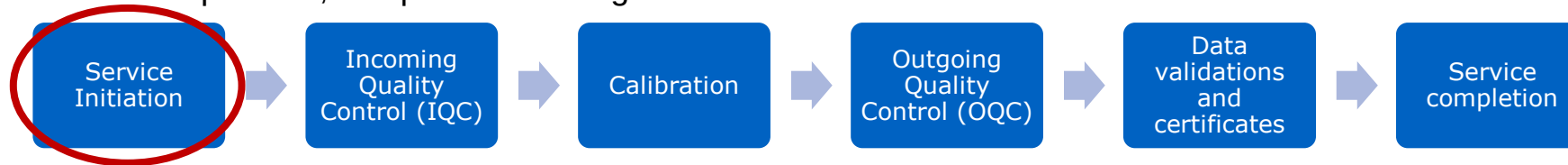
- These challenges put enormous workload to the first-line calibration engineers.
- The data is difficult to flow between different sites, between internal-external parties.

As the result, our calibration quality and efficiency are reduced by these challenges.

End-to-End Spectrometer Automatic Calibration Assistance Platform was proposed to overcome these difficulties.

2. Objectives

This is a typical calibration process, composed of 6 stages:



End-to-End Spectrometer **Automatic Calibration Assistance Platform** was aimed to improve the process.

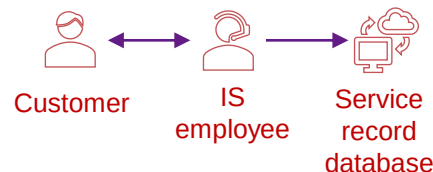
There are two keywords:

The **End-to-End** means we want to include the whole calibration process, from initiation to completion, from customer to customer, not just a few steps. Which means we cannot just introduce one or more tools, we need to try to improve the process as a whole.

The second keyword is **Automatic Calibration Assistance Platform**, not automatic platform. We don't aim to create fully automatic system to replace human. The whole process is still centered around human, the goal of the platform is to free our colleagues from tiresome, boring, repetitive works, so they can focus on high-value adding works.

We use **Service Initiation** for example:

The communication between the customers and our colleagues to finalize the details of upcoming calibration. The communication can happen in many forms and the needs varies. This is high-value adding works.

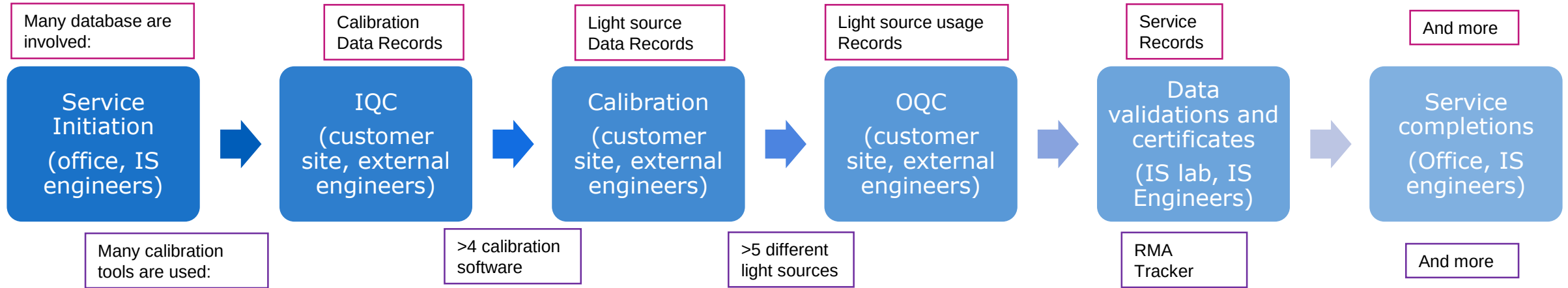


However, once negotiation is over, and when the customers send the calibration details in certain format, it is still up to the colleague to copy and paste one item at a time onto our service record database. This is boring and repetitive works.

The main objective of the platform is **to optimize and automate repetitive works, so people can focus on delivering outstanding high-value adding works. In order to improve quality and efficiency of whole calibration process.**

3. End-to-End Spectrometer Automatic Calibration Assistance Platform

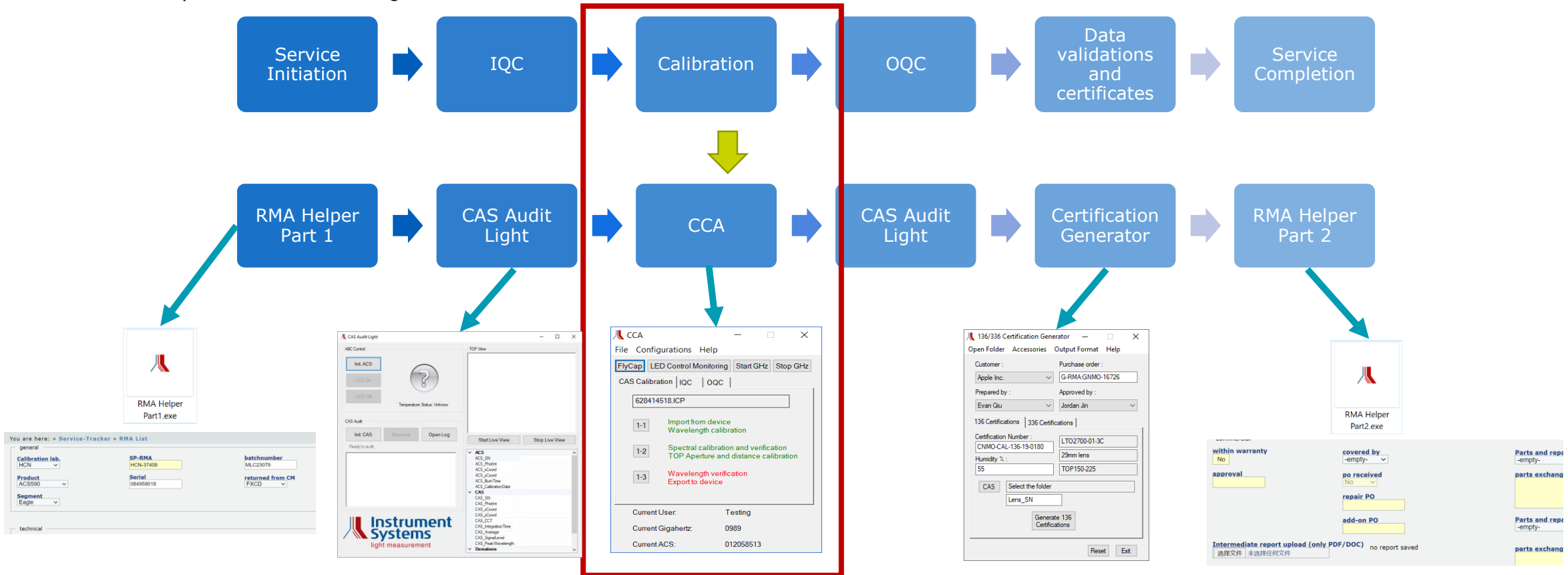
The whole calibration process is a very complicated process, both internal engineers and external engineers are involved in different location in many countries.



2+1 rules were followed throughout the design and improve of the platform:

1. If the action has no significant physical meaning, then automate it
Automate all software actions through Scripts, Marcos, API, SDK, ComOBJ, etc.
Automate insignificant physical interactions, like turn on/off power by electronic switch, etc.
 2. The exchange of data between softwares, locations, teams should be automatic with unified format
If the data needed is not new in the whole process, then the user shouldn't worry about it.
- +1.** Every automatic actions and intermediate data generated will be recorded and verified if possible.
The increase of efficiency will not decrease the quality, but also improving it.

By optimize and automate the whole calibration process, **5 master applications** were created for user to interact with. All software operations and data exchange are automatic performed in the background.



End-to-End Spectrometer **Automatic Calibration Assistance Platform** composed of these 5 applications along with improved workflow.

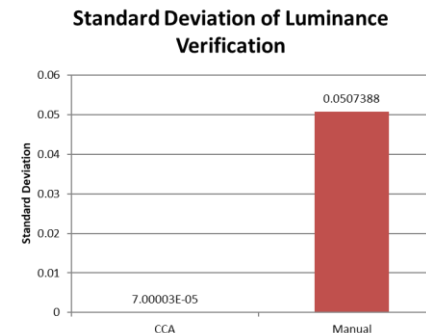
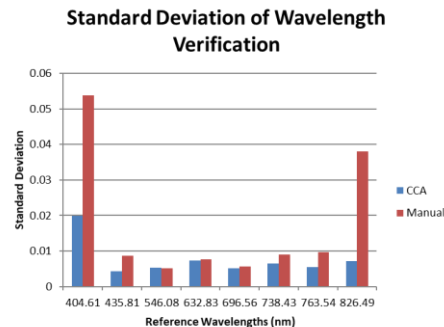
In order to verify the performance, we will take a closer look at **calibration stage** (the most complicated stage), and compare the results before and after the implantation of CCA.

4. Results comparison

To direct compare the calibration performance, we took the calibration data from our Hanoi site (Vietnam), and compared results one month before and one month after the deployment of CCA.

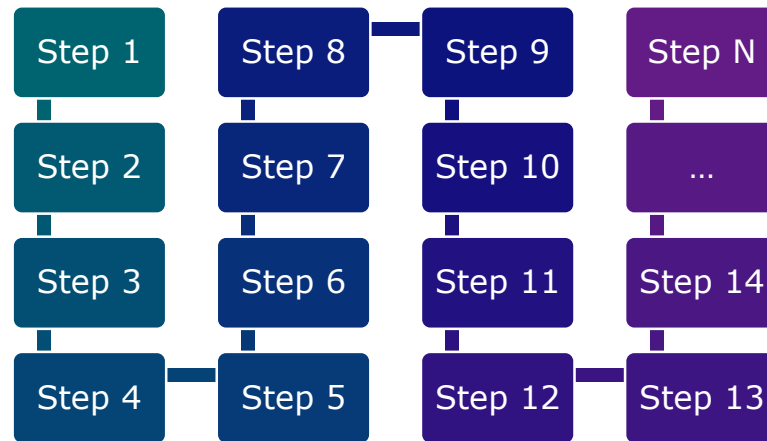
Before CCA	After CCA
30% calibrations records have structure errors (missing file, wrong file name, etc.)	0 error
31% calibrations data contain small errors (missing data, etc.)	0 error
One team of 2 engineers, calibrate 6 units a day Baseline (100%)	Two teams of 1 engineer, each calibrate 8 units half a day, total 16 units a day >266%*
The calibration accuracy is also increased, see charts below:	

*For 2 engineers to work full day, the estimated output would be 32 units, 533%



The improvements are significant, how is it achieved?

Before CCA

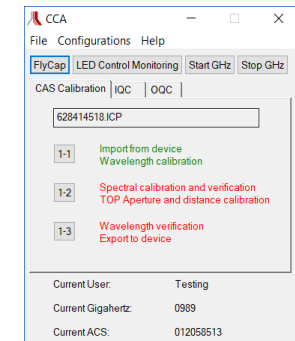


> 4 applications
> 5 databases
>100 operations

After CCA



1 master application (CCA)
No database interaction
3 operations



Interface of CCA

By optimize and automate the process, the operator's actions are reduced from >100 to just 3, and the operator doesn't need to worry about data exchange and recording.

Each step requires only 1 click of the button from operator.

5. Conclusions

End-to-End Spectrometer **Automatic Calibration Assistance Platform** is a successful project.

The company benefits from:

- >260% efficiency increase
- Reduced capital cost, e.g. calibration lamps have very short life, shorter work cycle = longer life span
- Reduced operational costs: e.g. hotel cost

And most important:

- Happier customers and Happier employees

Happy customer:

- Faster calibration cycle
- Flexible calibration cycle
- Increased equipment accuracy

Happy employees:

- Less boring, repetitive work
- Flexible work schedule
- Can focus on high-value adding works for better personal growth

Successful company, happy customer and employee composed a stable triangle.



KONICA MINOLTA

150

YEARS